ASSIGNMENT

DAX – PART 1 (THEORY)

1. What is DAX? Explain what *Data Analysis Expressions (DAX)* is and why it is useful in Power BI.

2. DAX Components Describe in your own words the difference between:

- Calculated Columns
- Measures
- Calculated Tables

3. DAX Function Types List the four main DAX function categories from Part 1:

- Aggregation
- Date/Time
- Logical
- Filter/Context

4. Why Learn DAX? Describe two benefits of using DAX when analyzing data in Power BI.

DAX – PART 1 (PRACTICAL TASKS)

1. Create a New Calculated Column – Customer Category Using the field `Customer[Age]`, create a DAX column with:

- Senior (Age ≥ 60)
- Adult (30–59)
- Youth (18–29)
- Minor (Age < 18)

2. Date Formatting Create a DAX calculated column transforming `Date[Date]` into:

`YYYY-MM`

3. Field Combination (NEW Task) Create a DAX column that combines:

`Customer[City]` and `Customer[Country]`

in the format:

`"City, Country"`

DAX – PART 1 (MEASURES + SUBMISSION)

1. Create a Basic Measure Write a measure that calculates total sales using:

```
Sales[Amount]
```

2. Identify Filters Explain how DAX evaluation context (row vs. filter context) affects calculated columns vs. measures.

Submit all answers as PDF or through Moodle.